1. Clinical Question & Core Hypothesis

Clinical Question

Tier-1 analysis highlights an apparent paradox—some Gleason grade group 5 tumors show dense lymphoid organization on histology/imaging yet still relapse early; why does local immune organization fail to provide durable tumor control in this highest-grade setting? 2) Tier-1 paths used: I built the hypothesis on (a) the CCL20–CCR6 immune-exhaustion axis (mechanistic route for immune suppression) and (b) DEK overexpression linked to shorter relapse; these together explain how intrinsic tumor programs might convert ostensibly immune-infiltrated niches into exhausted, ineffective immune microenvironments. I deprioritized the TCGA "subgroup" node as an anchor gene (kept it as a modifier) to preserve the 2-gene anchor limit and to keep the statement mechanistically specific. 3) Tier-2 support: cribriform and ductal/intraductal morphologies, single-cell/gene-expression programs, and predictive molecular biomarker signals support heterogeneity in immune composition and in tumor-intrinsic programs (they justify testing whether histologic patterns or molecular subtypes co-segregate with the DEK–CCL20 immune-suppressive state). 4) Endpoint choice: I selected PFI (progression-free interval/relapse-free survival) because the Ontologist emphasized DEK's link to relapse and PFI specifically captures early recurrence dynamics most relevant to aggressive Gleason 5 biology. 5) Deliberate exclusions: I excluded CCR6 from the anchor pair to avoid exceeding two anchors (placed CCR6 among secondary_modifiers); I also excluded broader exploratory morphology-only paths from the core claim to keep a tight mechanistic question (the exploratory paths remain for stratified analyses)?

Working Hypothesis

Among Gleason grade group 5 prostate tumors that exhibit TLS-like lymphoid aggregates, high tumor-cell DEK expression drives upregulation of the chemokine CCL20, creating a CCR6+ immune cell–enriched niche within TLSs. This DEK–CCL20 axis recruits/conditions CCR6+ populations (including T helper/effector subsets, regulatory cells, and immature dendritic cells) in a way that promotes T-cell exhaustion within the TLS microenvironment (increased PD-1, TIM-3, LAG-3 signaling), transforming TLSs from potentially protective structures into immunosuppressive hubs. Consequently, TLS-rich tumors with elevated DEK and CCL20 will exhibit impaired local anti-tumor activity and shorter progression-free interval (PFI) compared with TLS-rich tumors with low DEK/CCL20 or TLS-poor tumors. The effect is modulated by histologic patterns (cribriform vs ductal/intraductal features) and by additional transcriptional programs, but the core, testable mechanism centers on the DEK–CCL20 axis driving CCR6+ immune dysfunction within TLSs that correlates with early relapse.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	CCL20, DEK
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CCL20: CCL20–CCR6 immune-exhaustion axis (mechanistic route for immune suppression) and (b) DEK overexpression linked to shorter relapse; these together explain how intrinsic tumor programs might convert ostensibly immune-infiltrated niches into exhausted, ineffective immune microenvironments.

DEK: DEK overexpression linked to shorter relapse; these together explain how intrinsic tumor programs might convert ostensibly immune-infiltrated niches into exhausted, ineffective immune microenvironments.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CCR6	Supporting modifier from knowledge graph
Secondary	TCGA_subgroup	Supporting modifier from knowledge graph
Secondary	gene_expression_programs	Supporting modifier from knowledge graph
Secondary	cribriform_pattern	Supporting modifier from knowledge graph
Secondary	ductal_intraductal_features	Supporting modifier from knowledge graph
Secondary	molecular_biomarkers_predicting_response	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: with CCR6 as a functional mediator rather than a core anchor. Predictions and tests: - In TLS-positive Gleason 5 tumors, DEK and CCL20 co-expression will correlate with higher exhaustion marker expression (PD-1, TIM-3, LAG-

Prediction 2: in TLS-associated T cells and with immature/mature TLS signatures indicative of dysfunctional architecture. - CCR6+ cell density will be enriched in TLS-adjacent regions of DEK-high/CCL20-high tumors and will associate with shorter PFI. - Functional perturbation in model systems: upregulating DEK in prostate cancer cells will increase CCL20 secretion and suppress T-cell activation/exhaustion resolution in co-culture; blocking CCL20–CCR6 signaling will diminish TLS-associated T-cell exhaustion and restore anti-tumor activity in vitro and in vivo. - Stratification by morphologic patterns (cribriform vs intraductal) and ancillary gene-expression programs will reveal context-dependent modifiers but will not overturn the central DEK–CCL20–CCR6 mechanism.

Prediction 3: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CCR6: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- TCGA_subgroup: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.